Def. Let $(X, \mathcal{M}, \mu)$ be a measure space.

Define $L^1(\mu) \overset{\text{def}}{=} \{f : X \to \mathbb{C} \mid f \text{ is measurable with } \int_X |f| \, d\mu < \infty\}$

The members of $L^1(\mu)$ are called Lebesgue integrable functions.

Def. Let $f \in L^1(\mu)$ and $E \subset X$ be a measurable. Let $f = u + i \cdot v$.

Define $\int_E f \, d\mu \overset{\text{def}}{=} \int_E u^+ \, d\mu - \int_E u^- \, d\mu + i \cdot \left(\int_E v^+ \, d\mu - \int_E v^- \, d\mu \right)$

Thm. Suppose that $f, g \in L^1(\mu)$ and $\alpha, \beta \in \mathbb{C}$.

Then, $\alpha \cdot f + \beta \cdot g \in L^1(\mu)$ with

$$\int_X \alpha f + \beta g \, d\mu = \alpha \cdot \int_X f \, d\mu + \beta \cdot \int_X g \, d\mu.$$

Proof. Since the summation and product of measurable function is measurable, so $\alpha \cdot f + \beta \cdot g$ is measurable. And,

$$\int_X |\alpha \cdot f + \beta \cdot g| \, d\mu \leq \int_X |\alpha| \cdot |f| + |\beta| \cdot |g| \, d\mu = |\alpha| \cdot \int_X |f| \, d\mu + |\beta| \cdot \int_X |g| \, d\mu < \infty.$$

Thus $\alpha f + \beta \cdot g \in L^1(\mu)$.

Lebesgue's Dominated Convergence Theorem.

Thm. Suppose that $\{f_n\}$ is a sequence of complex measurable functions on X such that $f(x) = \lim_{n\to\infty} f_n(x)$ exists for every $x \in X$.

If there is a function $g \in L^1(\mu)$ such that: for all $n \in \mathbb{N}$,

$$|f_n(x)| \le g(x) \quad (x \in X)$$

then, $f \in L^1(\mu)$ with

$$\lim_{n\to\infty} \int_X |f_n - f|\, d\mu = 0 \quad \text{and} \quad \lim_{n\to\infty} \int_X f_n\, d\mu = \int_X f\, d\mu.$$

Proof. Since $|f| \le g$ and f is measurable, so $f \in L^1(\mu)$.

And, since $f_n(x) \le g(x)$ for all $n \in \mathbb{N}$, the limit $f(x) \le g(x)$. Therefore, the triangle inequality shows that $|f_n - f| \le 2g$.

Applying Fatou's Lemma to $2g - |f_n - f|$, then

$$\begin{aligned}
\int_X 2g\, d\mu &= \int_X \liminf_{n\to\infty} (2g - |f_n - f|)\, d\mu \\
&\le \liminf_{n\to\infty} \int_X 2g - |f_n - f|\, d\mu \qquad \text{Fatou} \\
&= \int_X 2g\, d\mu + \liminf_{n\to\infty} \left(-\int_X |f_n - f|\, d\mu\right) \qquad \int_X 2g\, d\mu \text{ is constant} \\
&= \int_X 2g\, d\mu - \limsup_{n\to\infty} \int_X |f_n - f|\, d\mu
\end{aligned}$$

Therefore, $\limsup_{n\to\infty} \int_X |f_n - f|\, d\mu = 0$, so that $\lim_{n\to\infty} \int_X |f_n - f|\, d\mu = 0$.